

Handout 9: AI-Powered Helpdesk Routing System

Business Scenario

Employees submit support requests in free-text format.

Examples:

Need leave for next Friday

VPN is not connecting

Travel reimbursement pending

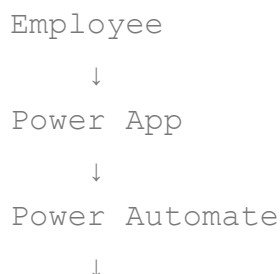
Need access to project folder

Laptop is not booting

The system should:

- Accept free-text requests
 - Classify the request using AI
 - Route the request to the correct department
 - Send notification to the responsible team
 - Notify the employee
-

Process Flow



AI Builder Prompt



Classification



Routing



Department Notification



Employee Confirmation

Power App Fields

Employee Name

Text Input

Employee Email

Text Input

Request Description

Multi-line Text

Examples:

Need leave next week

VPN is not connecting

Need reimbursement for hotel expenses

Laptop screen is broken

Submit Button

Power Automate Flow

Flow Name:

AI Helpdesk Routing Flow

Trigger:

Power Apps (V2)

Inputs:

EmployeeName

EmployeeEmail

RequestDescription

Step 1: Create AI Builder Prompt

Prompt Name:

Helpdesk Classifier

Prompt:

Classify the employee request into exactly one category.

Categories:

HR

IT

Finance

General

Rules:

HR:

Leave

Vacation

Attendance

Salary Slip

Employee Policy

IT:

Laptop

Computer

VPN

Outlook

Email

Password

Software

Finance:

Invoice

Expense

Travel Claim

Reimbursement

Payment

General:

Everything else

Return ONLY one value:

HR

IT

Finance

General

Request:

Input Variable:

RequestDescription

Step 2: Execute Prompt

Add:

Run a Prompt

Pass:

RequestDescription

Receive:

Category

Step 3: Routing Logic

Use:

Switch

Switch Value:

Category

HR Branch

Send Email:

hr@company.com

Subject:

New HR Request

IT Branch

Send Email:

`it@company.com`

Subject:

`New IT Request`

Finance Branch

Send Email:

`finance@company.com`

Subject:

`New Finance Request`

General Branch

Send Email:

`helpdesk@company.com`

Subject:

`General Support Request`

Step 4: Employee Confirmation

Send Email:

EmployeeEmail

Subject:

Request Submitted Successfully

Body:

Your request has been received.

Request:
[RequestDescription]

Assigned Category:
[Category]

The support team will contact you shortly.

Test Cases

Test Case 1

Need vacation next week

Expected:

HR

Test Case 2

My VPN is not connecting

Expected:

Test Case 3

Need reimbursement for hotel expenses

Expected:

Finance

Test Case 4

Need information about office cafeteria timings

Expected:

General

Challenge Enhancements

Enhancement 1

Add Priority Classification:

Low
Medium
High
Critical

Enhancement 2

Auto-create Ticket ID:

Enhancement 3

Send ticket to Microsoft Teams channel.

Enhancement 4

Store requests in:

Excel
SharePoint
Dataverse

Architecture Diagram

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graph TD; Employee --> Power_App[Power App]; Power_App --> Power_Automate[Power Automate]; Power_Automate --> AI_Builder_Prompt[AI Builder Prompt]; AI_Builder_Prompt --> Classification; Classification --> Switch_Routing[Switch Routing]; Switch_Routing --> Department_Team[Department Team]; Department_Team --> Employee_Notification[Employee Notification];
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